Dennis Ulmer

dennisulmer.eu · Kaleidophon · linkedin.com/in/d-ulmer/ · Dennis Ulmer

Employment

04.2025–today **Independent Identically Distributed Consulting (IID Consulting)** · Freelance Consultant *Amsterdam, NL*
- Focus: Independent research consultant helping clients research, develop and adopt specialized and tailored AI solutions, with, see iidconsulting.eu.
- Developed and shipped a finetuned language model for a specialized, challenging domain for an early-stage startup (details under NDA).

11.2024–today **Institute for Logic, Language and Computation (ILLC) / University of Amsterdam** · *Amsterdam, NL*
Postdoctoral Researcher
- Supervisor: Prof. Dr. Ivan Titov.
- Focus: Uncertainty quantification and communication in Natural Language Processing.

10.2023–02.2024 **Parameter Lab** · Research Intern *Tübingen, DE*
- Manager: Dr. Seong Joon Oh.
- Focus: Calibration for black-box large language models for question-answering in collaboration with researchers from Naver Labs, Seoul (KR).

06.2023–09.2023 **Amazon Web Services** · Applied Scientist Intern *Seattle, US*
- Managers: Dr. Elman Mansimov, Dr. Yi Zhang.
- Focus: Bootstrapping synthetic data for quick and flexible finetunung for task-oriented dialog.

02.2023–05.2023 **Higher Technical Institute (Instituto Superior Técnico)** · Visiting Researcher *Lisbon, PT*
- Supervisor: Prof. Dr. André F.T. Martins.
- Focus: Conformal prediction for token-level natural language generation .

06.2020–10.2020 **Pacmed** · Research Intern *Amsterdam, NL*
- Supervisor: Dr. Giovanni Cinà.
- Focus: Researched the limits of uncertainty estimation for neural networks for out-of-distribution detection on real-world medical tabular data.

12.2019–05.2020 **Institute for Logic, Language and Computation (ILLC) / University of Amsterdam** · *Amsterdam, NL*
Research Assistant
- Supervisor: Prof. Dr. Jelle Zuidema.
- Focus: Bridging the gap between human brain data and language model activations in the context of the ”Language in Interaction” project.

03.2017–08.2017 **MAJ // Digital** · Software Development Intern *Lisbon, PT*
- Supervisor: Paul Merz.
- Focus: Developing an NLP pipeline and graph database backend to extract a knowledge graph from Wikipedia / Wikidata, showing the involving of French politicians in political standards, using Python, Docker and Neo4j.

01.2016–01.2017 **Advanced Linguistic Technologies** · Student Developer *Weinheim, DE*
- Supervisor: Danny Rehl.
- Focus: Developing a Python backend for a proprietary RESTful-API for dynamic price adjustments in e-commerce.

Education

11.2020–05.2024 **IT University of Copenhagen (IT-Universitetet i København)** · PhD Studies *Copenhagen, DK*
- Supervisors: Dr. Christian Hardmeier (IT University of Copenhagen) and Dr. Jes Frellsen (Technical University of Denmark).
- Focus: Uncertainty Quantification for Natural Language Processing.
- Attended courses on advanced algorithms and data structures, Bayesian data analysis, graph theory.
- Dissertation: “On Uncertainty in Natural Language Processing” (🏆 runner-up to ACL dissertation award 2026).

09.2017–08.2019 **University of Amsterdam (Universiteit van Amsterdam)** · MSc Artificial Intelligence *Amsterdam, NL*

- Graduated *cum laude*, GPA 8.2/10.
- Attended courses on theoretical Machine Learning, Natural Language Processing, Computer Vision, Reinforcement Learning, Information Retrieval and other topics.
- Research on compositionality and incrementality in neural language models.
- Thesis: "Recoding latent sentence representations - Dynamic gradient-based activation modification in RNNs" under supervision of Dr. Dieuwke Hupkes and Dr. Elia Bruni.

10.2012–08.2016 **Heidelberg University (Ruprecht-Karls-Unversität Heidelberg)** · BA Computational Linguistics / BSc Geography *Heidelberg, DE*

- Graduated GPA 3.7/4 (1.6, *gut*).
- Attended courses on linguistics, programming, logic, statistics and machin learning as well as city geography, political geography and cartography.
- Thesis: "Analysis of relation prediction in German using word vector representations" under supervision of Dr. Yannick Versley and Dr. Vivi Nastase.

Publications

I have co-authored a total of 21 scientific publications, which according to Google Scholar per September 21, 2026 have accrued a total of 1,225 citations (h-index 15, i10-index 16). The following list excludes my dissertation, theses, and workshop proceedings.

*** = equal contribution**

Conference Papers

10.2026 **A. Kyriakou, D. Ulmer, I. Titov: *Shared Doubt: Zero-shot Cross-Lingual Confidence Estimation for Language Models*,** *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. *Budapest, HU*
PAPER CODE

08.2024 **D. Ulmer, M. Gubri, H. Lee, S. Yoon, S. Joon Oh: *Calibrating Large Language Models Using Their Generations Only*,** *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL) 2024*. *Bangkok, TH*
PAPER CODE ARTIFACTS

08.2024 **M. Gubri, D. Ulmer, H. Lee, S. Yoon, S. Joon Oh: *TRAP: Targeted Random Adversarial Prompt Honeypot for Black-Box Identification*,** *Findings of the Association for Computational Linguistics (ACL) 2024*. *Bangkok, TH*
PAPER CODE

08.2024 **D. Ulmer, E. Mansimov, K. Lin, J. Sun, X. Gao, Y. Zhang: *Bootstrapping LLM-based Task-Oriented Dialogue Agents via Self-Talk*,** *Findings of the Association for Computational Linguistics (ACL) 2024*. *Bangkok, TH*
PAPER

05.2024 **A. Farinhas, C. Zerva, D. Ulmer, A. Martíns: *Non-Exchangeable Conformal Risk Control*,** *International Conference on Learning Representations (ICLR) 2024*. *Vienna, AT*
PAPER CODE

03.2024 **D. Ulmer, C. Zerva, A. Martíns: *Non-Exchangeable Conformal Language Generation with Nearest Neighbors*,** *Findings of the European Chapter of the Association for Computational Linguistics (EACL) 2024*. *Valetta, MT*
PAPER CODE

12.2022 **D. Ulmer, J. Frellsen, C. Hardmeier: *Exploring Predictive Uncertainty and Calibration in NLP: A Study on the Impact on Method & Data Scarcity*,** *Findings of 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. *Abu Dhabi, AE*
PAPER CODE

12.2022 **D. Ulmer, E. Bassignana, M. Müller-Eberstein, D. Varab, M. Zhang, R. van der Goot, C. Hardmeier, B. Plank: *Experimental Standards for Deep Learning in Natural Language Processing Research*,** *Findings of 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. *Abu Dhabi, AE*
PAPER ARTIFACTS

07.2021 **D. Ulmer, G. Ciná. *Know Your Limits: Uncertainty Estimation with ReLU Classifiers Fails at Reliable OOD Detection*.,** Uncertainty in Artificial Intelligence (UAI) 2021 *Remote*
PAPER CODE

Journal Articles

08.2026 **S. Weber, A. Wang, A. Gupta, A. Subramonian, D. Ulmer, E. Tanwar, G. Aich, H. Devinney, J. Hobbs, J. Mickel, J. Tint, M. Sosto, R. Groshan, S. Astarita, V. Gautam, V. Blaschke, W. Agnew, W. Y. Lee, Y. Long: *Queer NLP: A Critical Survey on Literature Gaps, Biases and Trends*,** *Transactions of the Association for Computational Linguistics (TACL)* PAPER ARTIFACTS

06.2026 **D. Ulmer, A. Lorson, I. Titov, C. Hardmeier: *Anthropomimetic Uncertainty: What Verbalized Uncertainty in Language Models is Missing*,** *Transactions of the Association for Computational Linguistics (TACL)* PAPER CODE

10.2023 **D. Hupkes, M. Giulianelli, V. Dankers, M. Artetxe, Y. Elazar, T. Pimentel, C. Christodoulopoulos, K. Lasri, N. Saphra, A. Sinclair, D. Ulmer, F. Schottmann, K. Batsuren, K. Sun, K. Sinha, L. Khalatbari, R. Frieske, R. Cotterell, Z. Jin: *State-of-the-art Generalisation Research in NLP: A Taxonomy and Review*,** *Nature Machine Intelligence.* PAPER

04.2023 **D. Ulmer, C. Hardmeier, J. Frellsen: *Prior and Posterior Networks: A Survey on Evidential Deep Learning Methods For Uncertainty Estimation*,** *Transactions on Machine Learning Research (TMLR).* PAPER CODE

Workshop Papers

10.2026 **I. Brink, A. Boer, D. Ulmer: *Probing for Knowledge Attribution in Large Language Models*,** *The Ninth Workshop on Analyzing and Interpreting Neural Networks for NLP (BlackboxNLP).* *Budapest, HU* PAPER CODE

10.2026 **Y. Xia, D. Ulmer, T. Blevins, Y. Liu, H. Schütze, B. Roth: *Calibration Is Not Enough: Evaluating Confidence Estimation Under Language Variations*,** *Third Workshop on Uncertainty-Aware NLP (UncertaiNLP).* *Budapest, HU* PAPER

04.2022 **D. Ulmer, C. Hardmeier, J. Frellsen: *deep-significance: Easy and Meaningful Signifcance Testing in the Age of Neural Networks*.,** Machine Learning Evaluation Standards Workshop at ICLR 2022 *Remote* PAPER CODE

04.2022 **D. Ulmer, E. Bassignana, M. Müller-Eberstein, D. Varab, M. Zhang, C. Hardmeier, B. Plank: *Experimental Standards for Deep Learning Research: A Natural Language Processing Perspective*.,** Machine Learning Evaluation Standards Workshop at ICLR 2022 *Remote* PAPER CODE

12.2020 **D. Ulmer, L. Meijerink, G. Ciná. *Trust Issues: Uncertainty Estimation Does Not Enable Reliable OOD Detection On Medical Tabular Data*.,** Machine Learning for Health Workshop (ML4H) at NeurIPS 2020 *Vancouver, CA* PAPER CODE

08.2019 **J. Baan*, J. Leible*, M. Nikolaus*, D. Rau*, D. Ulmer*, T. Baumgärtner, D. Hupkes and E. Bruni. *On the Realization of Compositionality in Neural Networks*.,** 2nd Edition of the BlackboxNLP Workshop (BlackboxNLP) *Florence, IT* PAPER

08.2019 **D. Ulmer, D. Hupkes and E. Bruni. *Assessing incrementality in sequence-to-sequence models*.,** 4th Workshop on Representation Learning for NLP (Repl4NLP) *Florence, IT* PAPER

Pre-prints

07.2023 **J. Baan*, N. Daheim*, E. Ilia*, D. Ulmer*, H.-S. Li, R. Fernández, B. Plank, R. Sennrich, C. Zerva, W. Aziz: *Uncertainty in Natural Language Generation: From Theory to Applications*,** *Preprint.* PAPER

Service

Reviewing

- Conferences: ACL Rolling Review (*ACL, EMNLP), EurIPS (workshop proposals), NoDaLiDa, ICLR
- Journals: Transactions on Machine Learning Research (TMLR), Journal of Artificial Intelligence Research (JAIR).

- Workshops: Machine Learning Evaluations Workshop at ICLR (🏆 Outstanding reviewer), ACL Student Research Workshop, Blackbox NLP, UncertaiNLP, Queer in AI NeurIPS Workshop, Machine Learning for Health Workshop

Workshop Organization

2023–2024
- Co-founder and co-organizer of UncertaiNLP: First Workshop on Uncertainty-aware NLP, co-located with EACL 2024 in Valletta, MT.

Awards & Achievements

2026 🏆 Runner-up to the 2026 dissertation award by the Association for Computational Linguistis (ACL) for my dissertation"On Uncertainty in Natural Language Processing" among 38 other worldwide submissions.

2025 🏆 Awarded Cohere For AI Research Grant, consisting of 5000 USD in compute credits.

2022 🏆 Outstanding paper award for *Experimental Standards for Deep Learning Research: A Natural Language Processing perspective* at Machine Learning Evaluation Standards Workshop at ICLR 2022.

2022–today Developed open-source package `deep-significance` for statistical hypothesis testing for neural networks, starred by over 340 users on GitHub, downloaded over 59k times from PyPI (Python Package Index; determined via `pepy.tech`), and adopted in at least 60 academic publications (September 2026).

Teaching

11.2024-today **University of Amsterdam** · Guest Lecturer *Amsterdam, NL*
Guest lecturer for MSc Artificial Intelligence course "Natural Language Processing 2".

11.2020-05.2024 **IT University of Copenhagen** · Guest Lecturer *Copenhagen, DK*
Guest lecturer for MSc Data Science courses "Advanced Applied Statistics and Multivariate Calculus" and "Natural Language Processing 2".

09.2022-12.2022 **IT University of Copenhagen** · Teaching Assistant *Copenhagen, DK*
Providing practical exercise sessions and assistance on weekly assignments for MSc Data Science course "Advanced Applied Statistics and Multivariate Calculus".

02.2020-04.2020 **University of Amsterdam** · Teaching Assistant *Amsterdam, NL*
Supervising students on research projects on multilingual NLP for the MSc Artifical Intelligence course "Natural Language Processing 2".

09.2019-01.2020 **University of Amsterdam, Amsterdam University College** · Personal Tutor *Amsterdam, NL*
Tutored students on the MSc Artificial Intelligence course "Machine Learning 1" using Bishop's "Machine Learning & Pattern Recognition" as well as on BSc course "Natural Language Processing".

Supervision

MSc Theses

2025
- Ivo Brink (University of Amsterdam, NL): Provided day-to-day co-supervision for thesis "Real-time Knowledge Attribution as an Early-Warning Signal for LLM Hallucinations" (published at Blackbox NLP workshop 2026).

2025
- Athina Kyriakou (University of Edinburgh, UK): Provided day-to-day co-supervision for thesis "Shared Doubt: Probing for Language-Agnostic Uncertainty Features in Multilingual LLMs" (published at EMNLP 2026 main).

2025
- Eli Joe (University of Edinburgh, UK): Provided day-to-day co-supervision for thesis "Leveraging Chain-of-thought for Black-Box Uncertainty Quantification".

Talks

03.2024	**Technical University of Munich,** Title: Uncertainty Quantification for Natural Language Processing	*Munich, DE*
11.2023	**University of Stuttgart,** Title: Uncertainty Quantification for Natural Language Processing	*Stuttgart, DE*
02.2023	**University of Helsinki,** Title: Uncertainty Quantification for NLP: Current State & Open Questions	*Remote*
10.2022	**Industry Reading Group,** Title: Uncertainty Estimation in Natural Language Processing	*New York, US*
08.2019	**2nd Edition of the BlackboxNLP Workshop (BlackboxNLP),** Title: On the Realization of Compositionality in Neural Networks	*Florence, IT*
01.2019	**Computational Linguistics in the Netherlands (CLIN) 29,** Title: An Incremental Encoder for Sequence-to-Sequence Modelling	*Groningen, NL*

Community Building & Leadership

2025–today **Queer in AI** · Mentor, Organizer *Remote*
Mentoring students on PhD applications; organizing conference meetups (EMNLP 2025 in Suzhou, CN).

2021–2024 **Beers with NLPeers** · Founder, Organizer *Copenhagen, DK*
Initiator and organizer of recurring meet-up for researchers conducting Natural Language Processing research across five different universities in Copenhagen.

2014–2016 **Student parliament (StuRa) of Heidelberg University** · Elected Student Parliament Member *Heidelberg, DE*
Represented institute of Computational Linguistics in the student parliament and participarted in monthly plenary sessions and discussions..

2014–2016 **Department for Computational Linguistics, University Heidelberg** · Elected Member of Department Student Council *Heidelberg, DE*
Organized department-wide events, mediated communication between student body and faculty. Spearheaded an update to the curriculum of bachelor's program.